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Integrity, Wellbeing, and Administrative Capacity

An Analytics Assessment of Sanction Pathways and Service
Bottlenecks

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Executive Summary

This report evaluates how Ministry of Social Development's Integrity settings (sanction enforcement) interact with client Wellbeing and operational capacity, using a synthetic, MSD-like administrative dataset. Results point to a systemic "Northland bottleneck," where high enforcement intensity coincides with acute wellbeing stress and administrative gridlock. Overall benefit duration has a median of ~50.6 weeks and a heavy right tail, indicating that a small minority of long-duration cases disproportionately drive system reliance and should be targeted differently from the typical client pathway.

Northland stands out operationally: sanction exposure is highest (65.8 sanctioned clients per 100 clients). Within Northland's sanctioned cohort, clients with any Grade 3 sanction are strongly associated with subsequent food hardship demand (odds ratio 11.5 for a food grant within 60 days), suggesting severe sanctions may function as a crisis trigger in this context rather than a pathway to independence. This risk is plausibly amplified by throughput constraints: 53% of sanction-related actions remain open beyond 30 days.

The report recommends shifting from automated penalties to intensive, early case management for high-risk "Complex Families," and implementing a 30-day clearing rule to reduce backlog and prevent avoidable hardship escalation while protecting longer-term Independence.

AI and Synthetic Data Disclosure: This report uses a synthetic administrative-style dataset generated with assistance from Claude AI. Claude was used solely to simulate realistic, relational welfare data structures (clients, sanctions, hardship grants, and case notes) based on a high-level specification provided by the author (Appendix A). The synthetic data does not represent real MSD clients or outcomes. Its purpose is to demonstrate analytical reasoning, workflow design, and decision-support logic within an academic context.

Table of Contents

EXECUTIVE SUMMARY	1
1 BUSINESS CONTEXT AND ANALYTICAL PERSPECTIVE	1
2 DATA DESCRIPTION & METHODOLOGY	1
2.1 DATASET OVERVIEW.....	1
2.2 METHODOLOGICAL APPROACH	2
3 EXPLORATORY DATA ANALYSIS	2
3.1 INDEPENDENCE BASELINE: CONTINUOUS DURATION ON BENEFIT	2
3.2 REGIONAL DURATION COMPARISON: WHERE IS LONG-TERM DEPENDENCE CONCENTRATED?....	4
4 TARGETED ANALYTICAL TECHNIQUES.....	5
4.1 THE INTEGRITY PRESSURE POINT: SANCTION HOTSPOTS.....	5
4.2 THE WELLBEING SIGNAL: LINKING SANCTIONS TO HARDSHIP	7
4.3 THE BOTTLENECK EVIDENCE: ADMINISTRATIVE GRIDLOCK	9
4.4 CLIENT SEGMENTATION: WHO IS MOST AT RISK IN THIS BOTTLENECK?	11
5 BIAS, LIMITATIONS, AND MITIGATION	12
5.1 CORRELATION VS CAUSATION: WHAT CAN AND CANNOT BE CONCLUDED	12
5.2 SYNTHETIC DATA.....	13
5.3 ETHICAL IMPLICATIONS: SEGMENTATION MUST ADD SUPPORT, NOT PUNISHMENT	13
6 CONCLUSION AND RECOMMENDATIONS	13
6.1 WHAT THE EVIDENCE SUGGESTS	13
6.2 RECOMMENDATIONS	14
6.2.1 Pilot intensive supportive case management for “Complex families” following a severe breach (Northland).....	14
6.2.2 Introduce a 30-day clearing rule for sanction-related open actions (Northland)....	14
6.2.3 Add a balanced KPI: hardship escalation rate post-sanction.....	14
6.3 NEXT STEPS.....	14
APPENDIX: AI PROMPT FOR SYNTHETIC DATASET	15

1 Business Context and Analytical Perspective

Ministry of Social Development (MSD) delivers income support within a three-pillar mandate:

- Integrity: enforce eligibility rules and obligations
- Wellbeing: ensure clients can meet basic living needs
- Independence: support sustainable transitions into employment

While sanctions are designed as an integrity mechanism that encourages compliance, MSD leadership has raised a practical concern: in high-need settings, sanctions may not function as a “nudge to independence,” but as a trigger for crisis demand (e.g., hardship grants for essentials), while simultaneously increasing administrative workload for frontline teams.

From a Business Analyst perspective, the core issue is not whether sanctions exist, but how sanctions interact with operational processes and client vulnerability. This report therefore asks:

Do strict integrity mechanisms in high-need regions create administrative bottlenecks that harm client wellbeing and undermine pathways to independence?

This is a service design question as much as a policy question. If integrity actions consistently correlate with hardship escalation and unresolved case actions, the process may be generating work that prevents staff from doing the very support work required to move clients toward independence.

2 Data Description & Methodology

2.1 Dataset Overview

This analysis uses a synthetic administrative dataset designed to reflect MSD-like operational data structures while embedding a detectable bottleneck pattern for testing. The dataset contains four relational tables in Third Normal Form (3NF), linked by ClientID:

- Clients (2,500 records): demographics, region, benefit type, entry/exit dates, dependents, CaseComplexityScore
- Sanctions (2,284 records): sanction transactions, grade, payment suspension indicators, dates
- Hardship Grants (2,190 records): Emergency financial assistance, including grant type (e.g., food), amount, recoverability, dates
- Case Notes (4,915 records) Frontline process data capturing required actions, their resolution status (Open/Resolved), and dates.

Because this dataset is synthetic, the goal is not to claim real-world causality, but to demonstrate a decision-ready analytical narrative and surface the kinds of questions MSD would validate with real operational data.

2.2 Methodological Approach

The analytical workflow follows the Knowledge Discovery in Databases process:

1. Problem formulation (integrity–wellbeing–capacity interaction)
2. Data preparation and validation (date handling, joins, “as-of” window)
3. Exploratory analysis (duration distribution, regional baselines)
4. Targeted statistical and pattern analyses (sanction hotspots → hardship link → backlog evidence → segmentation)
5. Interpretation and recommendations.

A guiding principle is to use techniques only when they contribute directly to decision-making. Each analysis answers what we did, why it matters, how it was done, and so what the result implies for operational design.

3 Exploratory Data Analysis

3.1 Independence Baseline: Continuous Duration on Benefit

Before diagnosing bottlenecks, MSD needs a baseline for the independence pillar: what does a typical benefit pathway look like, and how uneven are outcomes? Without this baseline, later findings about sanctions or hardship risk being interpreted without context.

Continuous duration on benefit is calculated in weeks using entry and exit dates, with a fixed as-of date (31 Dec 2024) for clients without exit dates. Longer durations indicate slower transitions off support, making duration a pragmatic proxy for pathway length.

Metric	Value
Mean	63.3
Median	50.6
Std dev	44.5
IQR	76.4

Table 1. Continuous duration summary (weeks)

Table 1 summarises the distribution and immediately signals a key issue: the duration pattern is not normal. The mean (63.3 weeks) exceeds the median (50.6 weeks), and dispersion is high (SD 44.5; IQR 76.4). This matters because MSD’s “average client” is not a reliable planning unit. Averages are pulled upward by a minority of long-duration cases, which can distort both performance reporting and service design.

Figures 1 and 2 support this interpretation: the distribution is right-skewed with visible high-end outliers. The decision implication is methodological and practical: median-based comparisons are more credible for describing typical pathways, while targeted strategies may be required for the smaller group driving extended reliance.

This baseline also reframes resource strategy. If prolonged reliance is concentrated among a minority, independence interventions should avoid blanket designs that over-serve clients who will likely exit without intensive support. The next step is therefore to examine whether pathway length differs meaningfully by region.

Figure 1. Distribution of Continuous Duration on Benefit (Weeks)

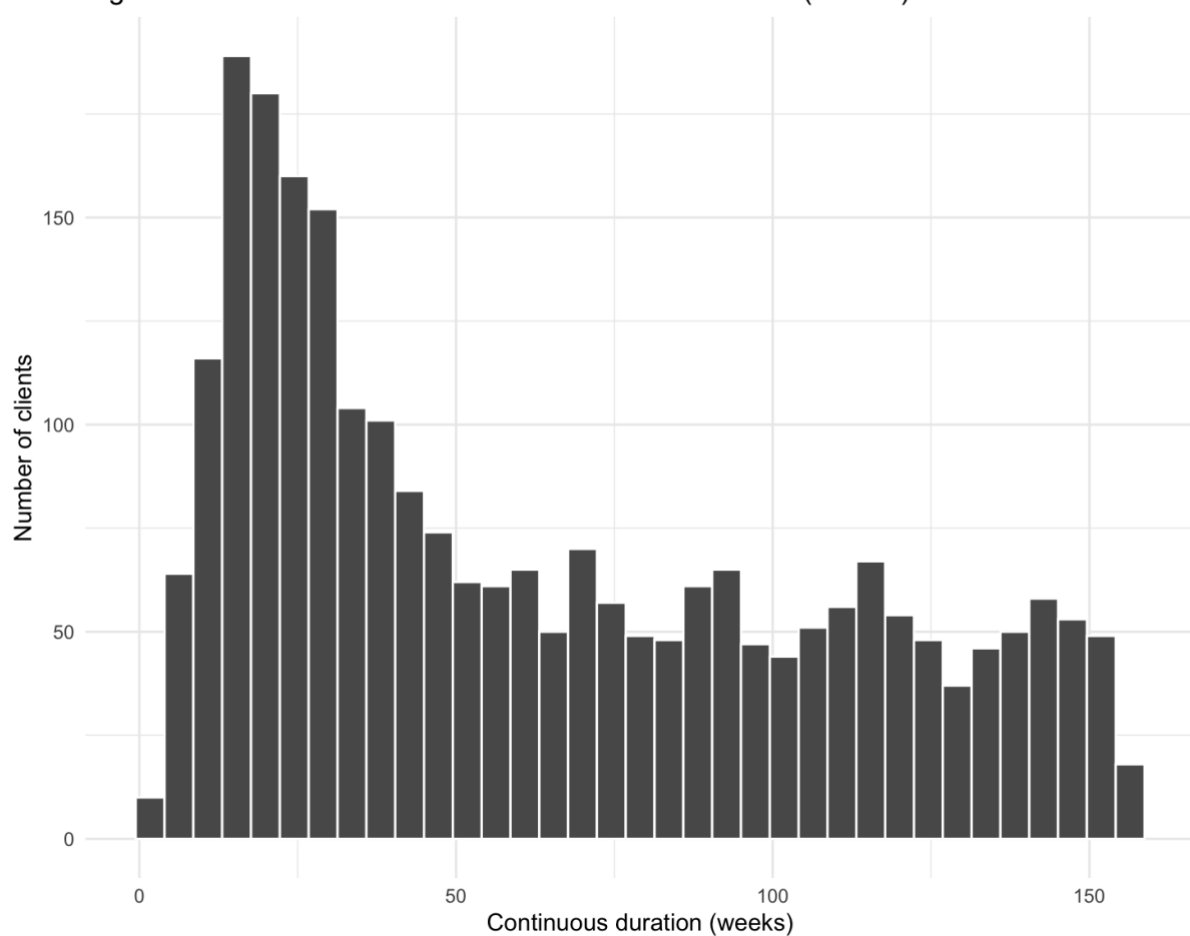


Figure 1. Distribution of continuous duration (histogram)

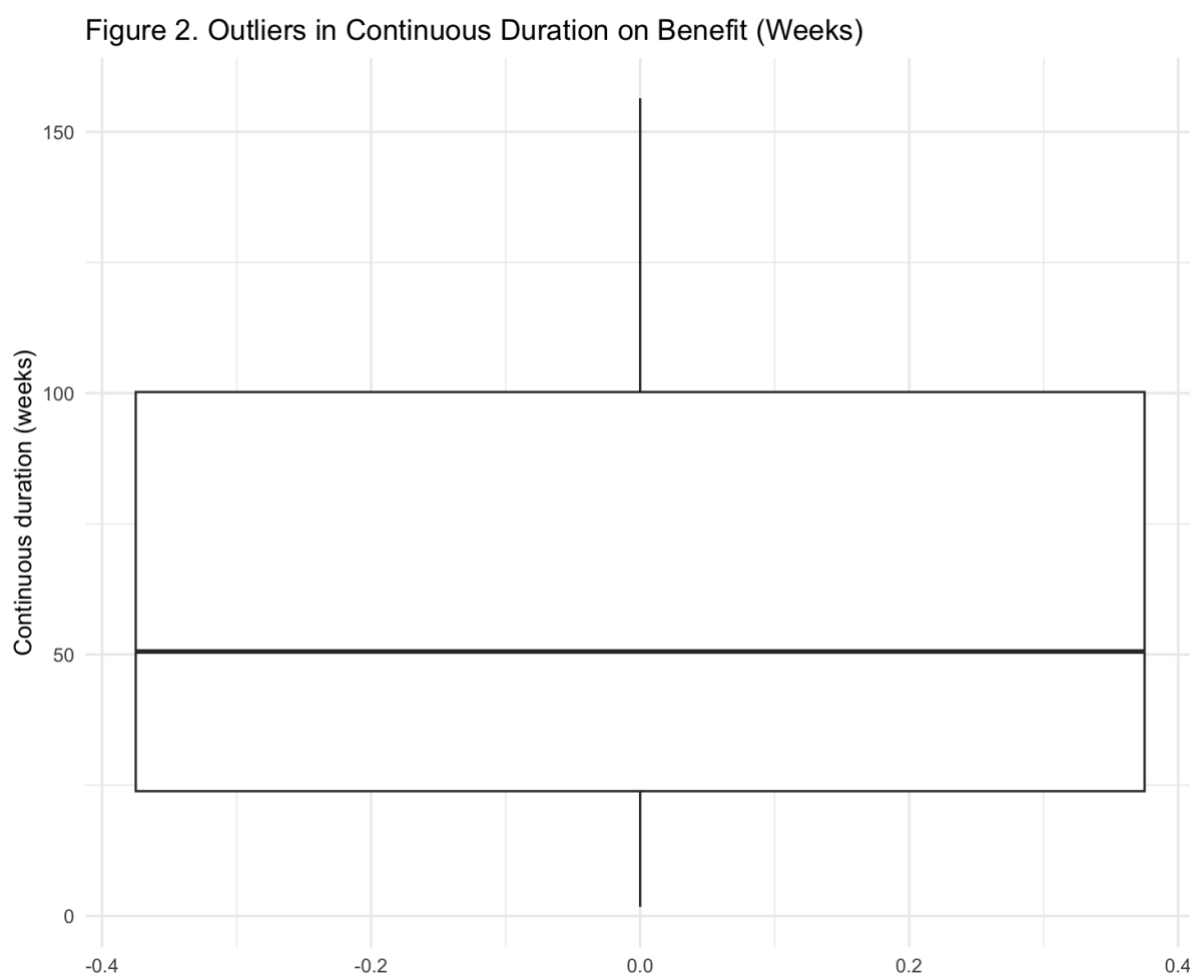


Figure 2. Outliers in Continuous Duration on Benefit (Weeks)

3.2 Regional Duration Comparison: Where Is Long-Term Dependence Concentrated?

Regional analysis helps MSD test whether some locations face consistently longer pathways, potentially reflecting labour market conditions, service access, or population need profiles. To balance “typical experience” and “long-tail risk,” the analysis compares median duration and 90th percentile (P90) duration across regions (Table 2; Figure 3). The median describes the typical client pathway, while P90 captures the upper tail where the most persistent cases sit.

Region	Clients	Median weeks	P90 weeks
Canterbury	413	56.7	134.0
Auckland	855	53.3	134.0
Bay of Plenty	254	50.4	138.0
Northland	494	47.9	134.0
Wellington	358	47.1	133.0
Other	126	37.4	124.0

Table 2. Duration by region (median and 90th percentile weeks)

Table 2 shows that Canterbury and Auckland have the highest median durations (56.7 and 53.3 weeks). Northland’s median is lower (47.9 weeks), and the P90 values are broadly similar across regions (~133-138 weeks). The key implication is not that duration is irrelevant, but that long-duration cases exist across all regions and the upper tail is not uniquely concentrated in one location. In other words, if operational strain or hardship escalation is concentrated in a particular region, duration alone may not explain it.

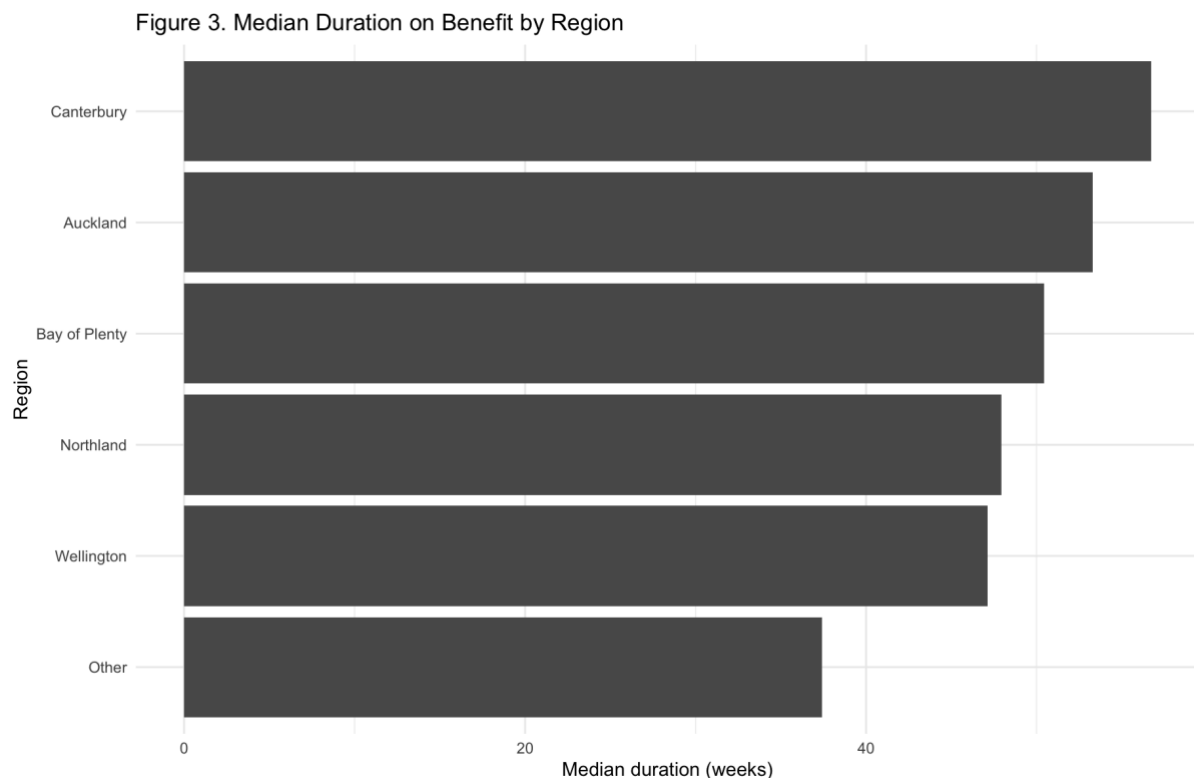


Figure 3. Median duration by region (context)

Figure 3 visualises this: the regional medians differ, but not in a way that would single-handedly account for major differences in workload or crisis demand. This pushes the analysis toward process dynamics. If outcomes are not radically different, MSD should ask whether process settings, particularly integrity mechanisms are creating uneven operational load and downstream effects. This leads into sanction exposure.

4 Targeted Analytical Techniques

4.1 The Integrity Pressure Point: Sanction Hotspots

The next question is whether sanctions are applied uniformly or whether enforcement intensity is uneven across regions. This matters because uneven intensity creates uneven administrative demand: sanctions are not just behavioural signals, they also trigger follow-up work (appointments, documentation checks, reviews, reinstatements, and re-engagement actions).

To avoid misleading comparisons across different population sizes, sanction exposure is measured as sanctioned clients per 100 clients (Table 3; Figure 4), and severity as % Grade 3 sanctions (Table 3; Figure 5).

Region	Clients	Sanctioned clients	Rate per 100 clients	Total sanctions	% Grade 3 (of sanctions)
Northland	494	325	65.8	621	24.3%
Wellington	358	156	43.6	300	22.7%
Canterbury	413	187	45.3	384	22.4%
Bay of Plenty	254	116	45.7	244	22.1%
Auckland	855	330	38.6	642	19.2%
Other	126	54	42.9	93	16.1%

Table 3. Sanction exposure and severity by region

Table 3 identifies Northland as an integrity hotspot: 65.8 sanctioned clients per 100 (highest of all regions) and a relatively high share of Grade 3 sanctions (24.3%). Interpreting this as simply “stricter practice” would be incomplete. A more decision-relevant interpretation is workload: higher sanction exposure means more compliance interactions and more resolution steps per client served. If local capacity is limited, this becomes a demand multiplier.

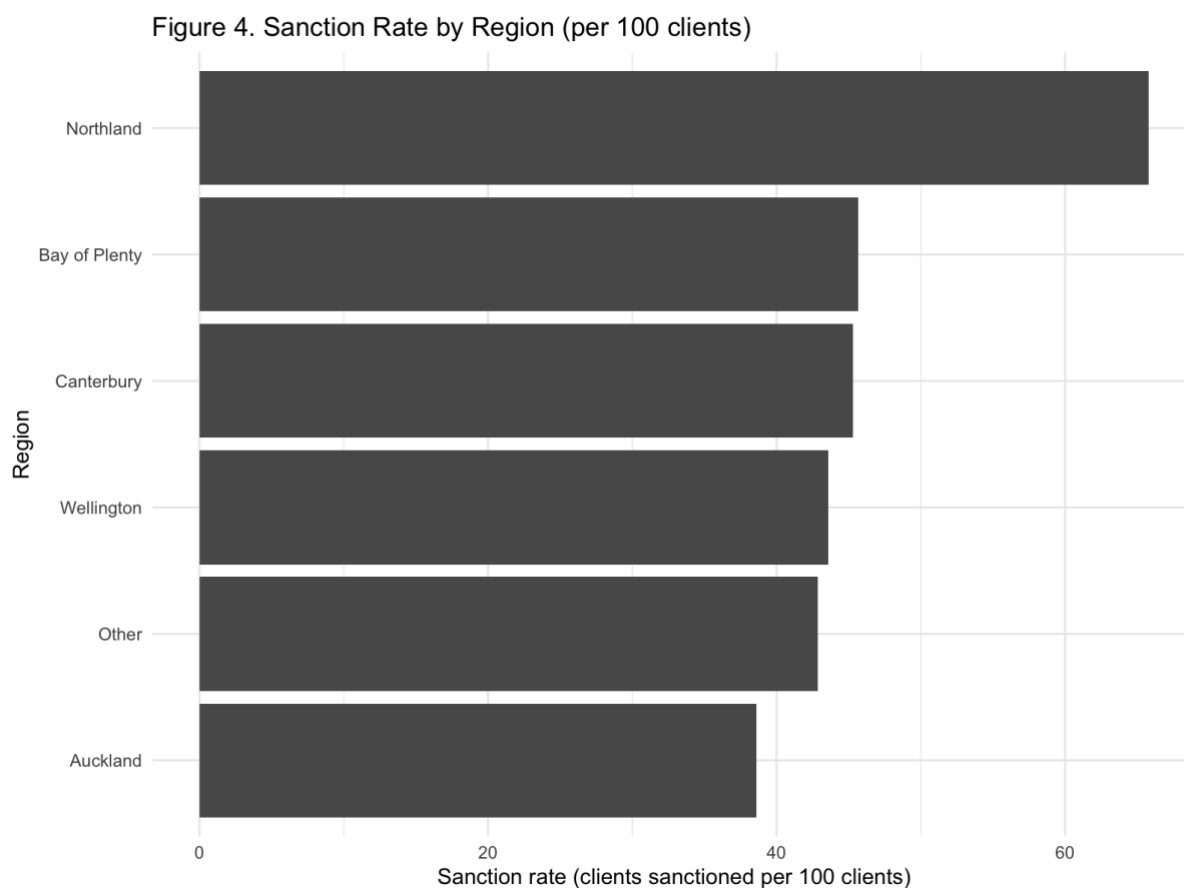


Figure 4. Sanction Rate by Region (per 100 Clients)

Figure 4 makes the hotspot pattern unmistakable: Northland’s sanction exposure sits clearly above all other regions. This supports the operational hypothesis that Northland is generating substantially more integrity-related workflow per client.

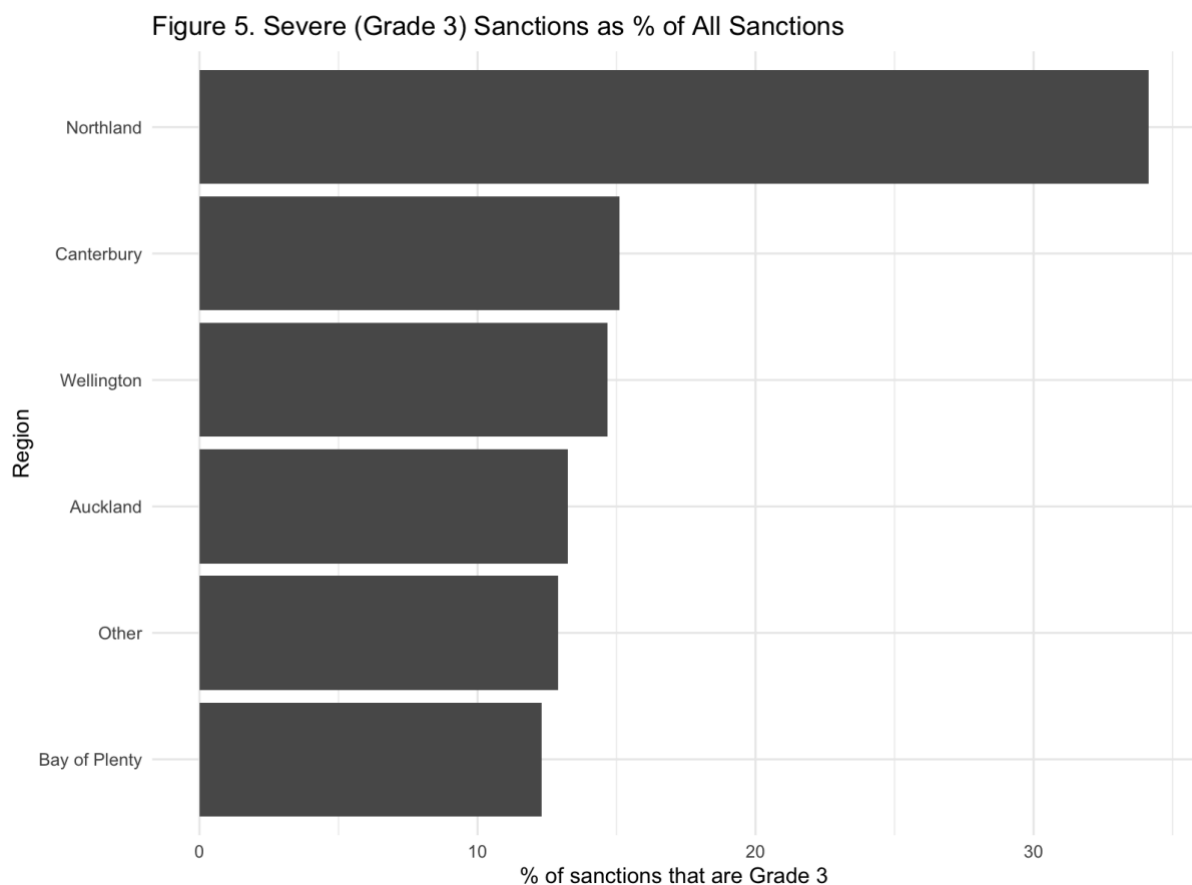


Figure 5. Severe (Grade 3) Sanctions as a Percentage of All Sanctions

Figure 5 adds why severity matters: if a region has both high exposure and relatively high severe sanctions, the potential for immediate income disruption rises. Severe sanctions typically involve payment suspension or significant reductions, increasing the risk of immediate financial distress. This combination of frequency and severity raises a critical wellbeing question: if severe sanctions are common, do we see an observable wellbeing signal shortly afterward?

4.2 The Wellbeing Signal: Linking Sanctions to Hardship

This section tests whether severe sanctions align with near-term wellbeing stress, using food hardship grants as a concrete indicator of acute need. A simple correlation is not appropriate because the outcome is binary (food grant within a defined time window). Instead, the analysis uses an event-anchored flag: a client is flagged if they receive a food hardship grant within 60 days after a Grade 3 sanction. The resulting regional rates are presented in Table 4 and Figure 6.

Region	% of all clients with food hardship ≤60 days after Grade 3 sanction
Northland	18.8%
Other	1.6%
Bay of Plenty	1.6%
Canterbury	1.5%
Auckland	0.9%
Wellington	0.8%

Table 4. Regional prevalence of food hardship within 60 days of a Grade 3 sanction

The decision-critical result is contrast, not merely magnitude. Northland’s post–Grade 3 food hardship signal is dramatically higher (18.8%) than all other regions (each below ~1.6%). This matters for governance because the same type of sanction does not show the same wellbeing pattern elsewhere. The implication is not “sanctions always cause hardship,” but that in Northland, sanctions may be interacting with local conditions (vulnerability, service access, or throughput delays) in a way that increases crisis demand.

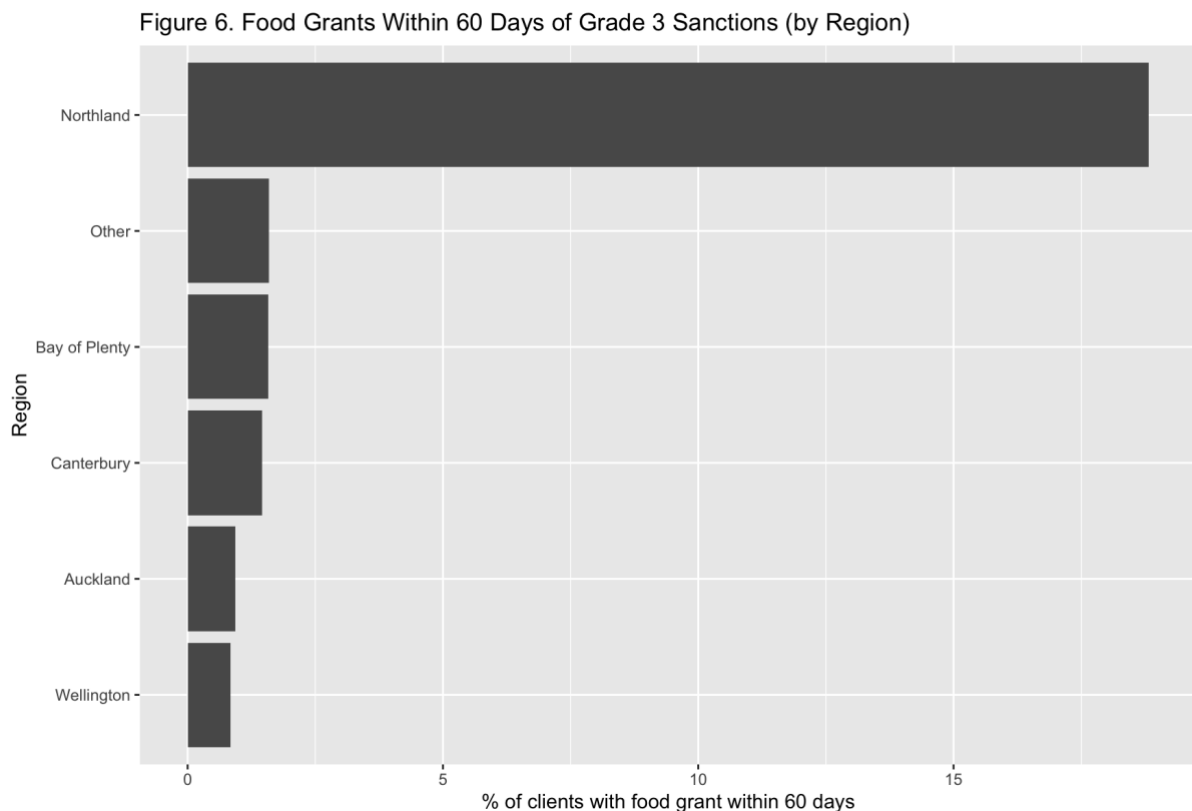


Figure 6. Food Hardship Grants Within 60 Days of Grade 3 Sanctions (by region)

Figure 6 therefore functions as a screening result. It identifies where the signal is strongest, but it does not isolate whether the signal is driven by sanction severity or by the vulnerability profile of sanctioned clients. To tighten decision relevance, the next step

restricts analysis to Northland’s sanctioned cohort and compares severity exposure within that cohort.

Within Northland, sanctioned clients are divided into two groups: those who have ever received a Grade 3 sanction and those who have been sanctioned but never at Grade 3. A contingency table compares these groups against receiving a food grant within 60 days (Table 5). An odds ratio quantifies effect strength.

Grade 3 sanction	Food grant ≤60 days	No food grant ≤60 days	Total	% with food grant
No	14	138	152	9.2%
Yes	94	79	173	54.3%
Total	108	217	325	—

Odds Ratio (Grade 3 vs No Grade 3): 11.54

95% Confidence Interval: 6.34 – 22.46

p-value (Fisher’s exact): < 0.001

Table 5. *Association Between Grade 3 Sanctions and Food Hardship Within 60 Days (Northland Sanctioned Clients)*

This finding represents a pivotal point in the analysis. Within Northland’s sanctioned population, Grade 3 exposure is associated with a substantially higher likelihood of near-term food hardship, with a large odds ratio and strong statistical significance.

The critical interpretation is not that sanctions cause hardship, but that within the enforcement pathway, severity is strongly associated with crisis support. In the operational reality, it raises a governance concern: integrity actions may be shifting costs toward hardship budgets and frontline workload, potentially undermining wellbeing and independence outcomes.

That creates the next operational question: why is the association so pronounced in Northland? Two plausible explanations compete:

- Vulnerability mechanism: the most vulnerable clients are both more likely to receive severe sanctions and more likely to need food support.
- Process mechanism: severe sanctions create follow-up work that cannot be resolved quickly, extending disruption and triggering hardship requests.

To test the process mechanism, the report examines backlog.

4.3 The Bottleneck Evidence: Administrative Gridlock

If sanctions generate administrative work that exceeds capacity, one observable symptom is aged open actions - case actions that remain unresolved long after they should have been completed. Using case notes for sanctioned clients, the analysis

calculates the percentage of actions that are Open and older than 30 days as of 31 Dec 2024 (Table 6; Figure 7).

Region	% open actions >30 days
Northland	53.0%
Canterbury	30.0%
Auckland	29.3%
Bay of Plenty	28.9%
Wellington	27.0%
Other	25.5%

Table 6. Administrative backlog by region (sanctioned clients)

Table 6 shows a striking gap: Northland has 53% open actions older than 30 days, compared to roughly 25–30% elsewhere. This is operationally significant. Aged open actions imply unresolved obligations, incomplete follow-ups, or delayed interventions. At scale, this means clients remain “stuck” while staff time is consumed by backlog rather than proactive progression.

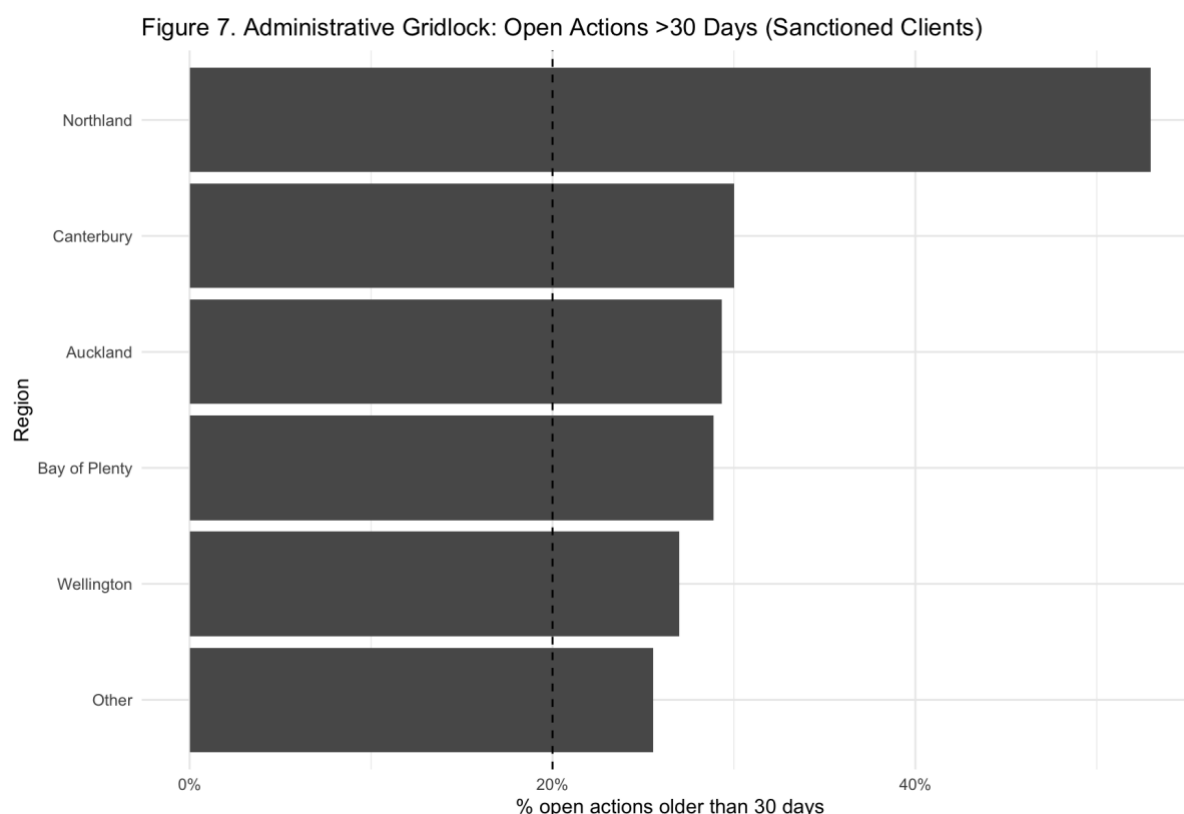


Figure 7. Administrative Gridlock: Open Sanction-Related Actions Over 30 Days

Figure 7 strengthens decision meaning through a reference threshold (20%). All regions exceed it, indicating system-wide strain, but Northland is far beyond it. This offers a plausible mechanism linking earlier results: if severe sanctions generate follow-up requirements and a large share of these remain open for long periods, clients may

experience longer disruption and higher crisis need, consistent with the food hardship signal.

This finding does not prove causality, but it strengthens the service design hypothesis: in Northland, integrity intensity appears misaligned with operational throughput, increasing the risk of a backlog-hardship feedback loop. The next question becomes: who is most exposed to harm when the system is congested?

4.4 Client Segmentation: Who Is Most at Risk in this Bottleneck?

If the system is congested, not all clients face equal risk. Uniform responses can be inefficient (over-servicing low-need clients) and harmful (under-supporting those most vulnerable to disruption and delay). Segmentation therefore shifts the analysis from diagnosis to service design.

Using Northland's sanctioned clients, k-means clustering (k=3) is performed on standardised values of:

- Age
- Dependents count
- Case complexity score
- Sanction count

Cluster profiles are summarised (Table 7) and visualised (Figure 8).

Cluster persona (interpretive label)	n	Avg age	Avg dependents	Avg complexity	Avg sanctions
Complex families (high dependents, high complexity)	70	30.3	3.37	3.99	~2.1
Long-term/older with moderate complexity	103	39.1	0.72	2.80	~2.0
Young solo jobseekers (lower dependents)	152	26.1	0.72	2.53	~2.0

Table 7. Cluster profiles (Northland sanctioned clients)

The decision value of clustering is practical: it supports targeted intervention design. One cluster stands out as the most vulnerable to backlog harm: clients with higher dependents and higher complexity ("Complex families"). Importantly, sanction exposure is similar across clusters, implying the differentiator is not sanction frequency but resilience to disruption and delay. Households with dependents are less able to absorb income shocks, and delays in resolution are more likely to translate into acute hardship, consistent with the observed food grant signal.

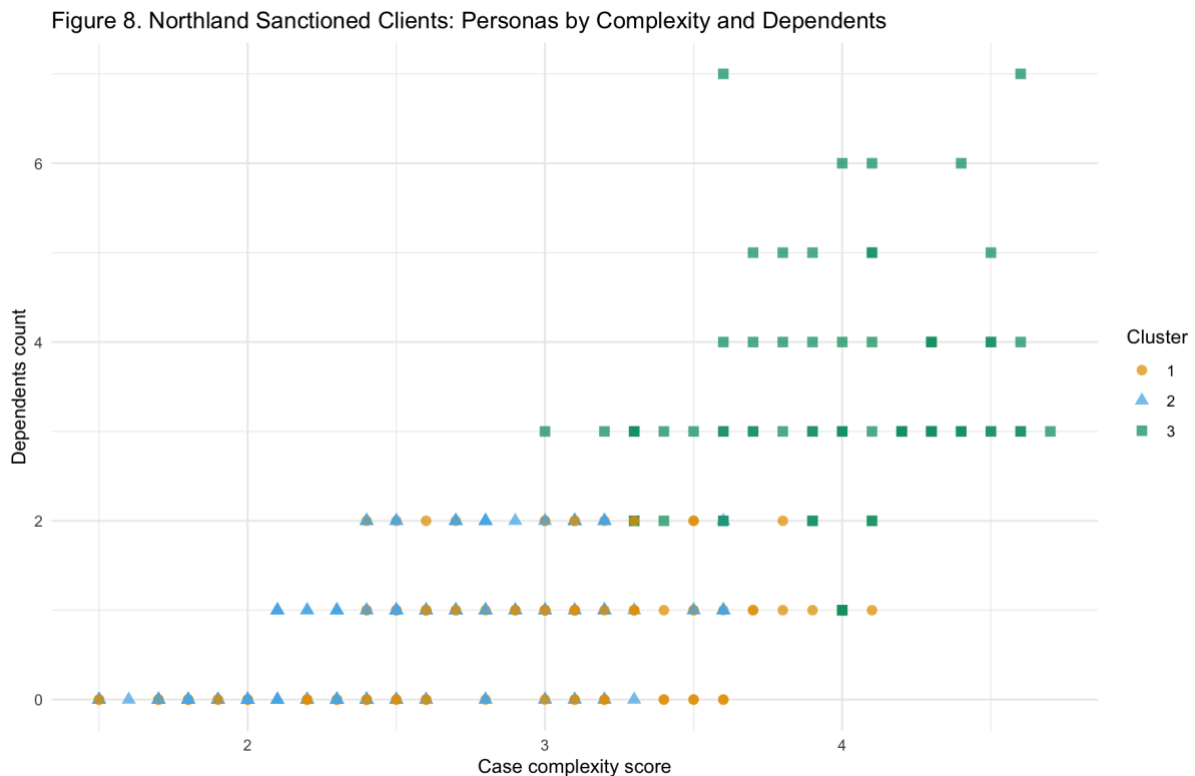


Figure 8. Northland Sanctioned Clients: Personas by Complexity and Dependents

Figure 8 makes this policy point concrete. The high-dependents/high-complexity group forms a visually distinct segment compared to younger solo jobseekers and older moderate-complexity clients. This supports a critical conclusion for decision-makers: a one-size-fits-all sanction pathway treats fundamentally different client realities as equivalent. If MSD wants to preserve integrity without generating avoidable hardship, differentiating the pathway for the most vulnerable group is a more defensible solution than removing sanctions entirely.

5 Bias, Limitations, and Mitigation

5.1 Correlation vs Causation: What Can and Cannot be Concluded

The strongest association (Grade 3 and food hardship in Northland) is not proof that sanctions cause hardship. Alternative explanations include:

- Underlying vulnerability: barriers may drive both severe sanctions and food insecurity.
- Practice variation: hardship uptake may reflect accessibility or referral norms rather than true need.
- Measurement choices: the 60-day window is plausible but not definitive; other windows may change effect size.

If validated with real MSD data, mitigation would include richer covariates (employment status, prior hardship, health flags), sensitivity checks across time windows, and interpretable causal-inference approaches (e.g., matching) where appropriate.

5.2 Synthetic Data

Because patterns are embedded, this report is best treated as a decision logic blueprint, not an evaluation of MSD performance. Its value lies in demonstrating an evidence chain: identify hotspots, detect wellbeing signals, test operational throughput, then design targeted interventions.

5.3 Ethical Implications: Segmentation Must Add Support, not Punishment

Segmentation can be misused to intensify surveillance or punitive targeting. This report adopts an ethical stance: segmentation should route clients to added support, not escalated enforcement. Success should be measured through reduced hardship escalation and faster resolution, not higher sanction throughput.

6 Conclusion and Recommendations

6.1 What the Evidence Suggests

This report established that independence outcomes are heavy-tailed and require careful interpretation. Regional duration analysis showed Northland is not the longest-duration region by median pathway length, meaning later Northland concerns should not be attributed to duration alone.

Northland becomes strategically important because three factors converge:

- Integrity intensity: highest sanction exposure (65.8 per 100 clients).
- Wellbeing shock signal: markedly higher post-Grade 3 food hardship signal, reinforced by within-Northland contingency evidence.
- Operational congestion: the highest backlog (53% open actions >30 days), offering a plausible mechanism for hardship escalation via delayed resolution.

The conclusion is not “sanctions fail.” It is:

In Northland, the interaction between sanction intensity and administrative capacity appears misaligned with wellbeing outcomes and may undermine independence by creating a loop of disruption, backlog, and crisis demand.

6.2 Recommendations

6.2.1 Pilot intensive supportive case management for “Complex families” following a severe breach (Northland).

- Trigger: first severe event (e.g., Grade 3 or defined severe breach).
- Routing: assign an intensive case manager within five working days.
- Rationale: this group is least able to absorb disruption and most exposed to harm when resolution is delayed.
- Measures: reduce post-sanction food hardship, reduce aged open actions, improve time-to-resolution and engagement.

6.2.2 Introduce a 30-day clearing rule for sanction-related open actions (Northland).

- Mechanism: actions open beyond 30 days enter triage/escalation for resolution.
- Rationale: backlog is the clearest throughput indicator; reducing aged open actions directly addresses bottleneck mechanics.
- Measures: reduce Northland open>30 rate toward $\leq 20\%$, track timeliness and workload indicators.

6.2.3 Add a balanced KPI: hardship escalation rate post-sanction.

- Metric: % of sanctioned clients receiving food hardship within 60 days after severe sanction (plus optional broader wellbeing indicator).
- Rationale: ensures integrity optimisation does not inadvertently undermine wellbeing.
- Governance: treat as harm-detection, reported alongside sanction volumes and resolution timeliness.

6.3 Next Steps

Before implementing policy change, MSD should validate using real administrative data:

1. Re-run analysis with richer vulnerability covariates.
2. Sensitivity checks (30/60/90 days; different grant types).
3. Map the sanction-to-resolution workflow in Northland to locate delay points.
4. Co-design the pilot with frontline teams to ensure feasibility and avoid burden-shifting.

Appendix: AI Prompt for Synthetic Dataset

I am a business analyst simulating administrative data for the New Zealand Ministry of Social Development (MSD). The purpose is to create a realistic, relational dataset to analyse whether strict enforcement of welfare obligations (called 'Integrity actions') in regions with systemic socio-economic disadvantage creates service bottlenecks. These bottlenecks are hypothesised to worsen client wellbeing (measured by increased urgent hardship grants) and prolong benefit dependence, undermining MSD's goals of client independence.

Generate a synthetic relational dataset with the following tables, ensuring records can be linked via a unique ClientID:

1. Clients Table: ClientID, Region, Age, Gender, Ethnicity, BenefitType (e.g., Jobseeker Support, Sole Parent Support), Dependents_Count, EntryDate, ExitDate (some NA for active), CaseComplexityScore (a score 1-5 you generate based on other variables, e.g., based on number of dependents, prior hardship events).
2. Sanctions Table: SanctionID, ClientID, SanctionDate, SanctionType (e.g., 'Work Test Failure', 'Appointment Non-Attendance'), SanctionGrade (1=Minor, 3=Severe), PaymentSuspended (Y/N), SuspensionDuration_Days.
3. Hardship_Grants Table: GrantID, ClientID, GrantDate, GrantType (e.g., 'Special Needs Grant - Food', 'Advance Payment'), Amount, Recoverable (Y/N).

Crucial Instruction for Data Patterns:

Inject the Bottleneck Signal: For clients in the 'Northland' region, create a strong pattern where receiving a Grade 3 sanction is frequently followed (within 4-8 weeks) by one or more 'Special Needs Grant - Food' hardship events. This correlation should be noticeably stronger in Northland than in other regions like 'Auckland' or 'Wellington'.

Create Process Data: Include a CaseNotes table with ClientID, NoteDate, ActionRequired (e.g., 'Follow-up on sanction', 'Assess hardship application'), ActionStatus ('Open', 'Closed'). Ensure Northland has a higher proportion of 'Open' actions for sanctioned clients, simulating administrative overload.

Avoid Simple Averages: Generate raw, instance-level data. Do not pre-aggregate.

Skew Distributions: Make ContinuousDurationWeeks and HardshipGrants_Count right-skewed, a majority with low counts/duration, but a long tail of clients with very high values.